IN THE RANDOM SUBGRAPH OF A HIGH-DIMENSIONAL PRODUCT GRAPH THERE IS TYPICALLY NO COMPONENT WHOSE ORDER LIES IN THE INTERVAL FROM TWO TO D TO THE TWENTIETH

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned author source sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The rooted-tree bound used in the proof and the displayed range for p are navigation pointers only; their content is declared as named standard prerequisites and they are not reproduced.

Allowed standard prerequisites (editorial).

- The parameters n, d, p and the constant M, the relation d = Theta_M(log n), and the low-degree expansion estimates of the product graph. Set up in the author's preliminaries and proof outline (source lines 201-240); the hypothesis n(1-p)^d <= d^{1/50} used at the end of the proof is the author's displayed eq:mu-range cited below as a pointer. Supplies the expansion bound e_G(K,K^c) >= k(d - (M-1) log_M k) applied to a set K of size k, and the parameter relations that turn the probability bound into n^{1-0.8k}.

Source reference. Original lemma 2.3: sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex, source lines 231–231. Navigation pointer only: the counted proof uses the rooted-tree bound in one sentence to control the number of spanning trees; that bound is declared as a named standard prerequisite and is not reproduced here.

Source reference. Original equation 4.2: sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex, source lines 479–479. Navigation pointer only: the counted proof uses the range of p displayed there in one step of the substitution; the display is delivered verbatim in this lane's card for lem:isolates-far but is not part of this card.

Lemma 0.1. *Whp, G_p has no component whose order lies in $[2, d^{20}]$.*

$$e_G(A, A^c) \geq \frac{e}{M}|A| \ln \frac{n}{|A|}, \tag{1}$$

$$e_G(A, A^c) \geq |A|\big(d - (M-1)\log_M |A|\big). \tag{2}$$

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Proof. Fix $2 \le k \le d^{20}$. For every set $K \subseteq V$ of size k, the expansion from (2) gives

$$e_G(K, K^c) \ge k\big(d - (M-1)\log_M k\big) \ge 0.9kd.$$

If K is a component of G_p, it contains a spanning tree all of whose edges are present, while every edge of $E_G(K, K^c)$ is absent. Thus, using Original lemma 2.3, we obtain

$$\mathbb{P}(G_p \text{ has a component of order } k) \le n(ed)^{k-1}(1-p)^{0.9kd} \le n^{1-0.8k},$$

Where the last inequality follows since $2 \le k \le d^{20}$, $n(1-p)^d \le d^{1/50}$ and $d = \Theta_M(\ln n)$. Summing over k shows that whp there is no component whose order lies in $[2, d^{20}]$. □